- Customer advices are issued in a timely manner.
- Any funds transfer requests received via telex, telephone or fax are authenticated prior to processing.

Objective 13: Determine if the institution complies with the Federal Reserve Policy Statement on Payments System Risk.

1. Determine if the institution incurs overdrafts in its Federal Reserve account. If so, consider if:

- The institution has reviewed and complied with the Payment System Risk program (i.e., the institution selected an appropriate net debit cap).
- The institution has elected a de minimis or self-assessed net debit cap and ensure that the examination evaluates the adequacy of records supporting the accuracy of the de minimis or self-assessed rating.

Objective 14: Review the institution's policies and procedures regarding the release of payment orders to assess the adequacy of controls.

1. Determine whether all incoming and outgoing payment orders and messages are received in the funds transfer area.

2. Obtain a sample of payment orders. Determine if the payment orders are:

- Logged as they enter the funds transfer department.
- Time stamped or sequentially numbered for control.
- Reviewed for signature authenticity.
- Reviewed for test verification, if applicable.
- Reviewed to determine whether personnel who initiated each funds transfer have the authority to do so.

3. Determine if current lists of authorized signatures are maintained in the wire transfer area. Ensure the lists indicate the amount of funds that individuals are authorized to release.

4. Assess whether there are adequate dual controls over the review of payment orders

and message requests. Determine whether an independent employee reviews the requests for the propriety of the transaction and for future dates, especially on multiple transaction requests.

Objective 15: Coordinate the review of wholesale payment systems with examiners in charge of reviewing other information technology risks.

1. In discussion with other examiners, ensure that management applies corporate-wide, information technology policies and procedures (i.e. development and acquisition, operational security, environmental controls, etc.) to the funds transfer department. If any discrepancies exist, determine their severity and document any corrective actions.

Appendix B: Glossary

Aggregate Short Position - The sum of a Settlement Member's short positions, each such short position expressed in its base currency equivalent and adjusted by the applicable haircut.

Aggregate Short Position Limit - In respect of a Settlement Member, the maximum aggregate short position that such Settlement Member is permitted to incur at any time.

Authentication - The process of verifying the identity of an individual user, machine, software component, or any other entity.

Automated Clearing House (ACH) - An electronic clearing system in which a data processing center handles payment orders that are exchanged among financial institutions, primarily via telecommunications networks. ACH systems process large volumes of individual payments electronically. Typical ACH payments include salaries, consumer and corporate bill payments, interest and dividend payments, and Social Security payments.

Automated Teller Machine (ATM) - An electronic funds transfer (EFT) terminal that allows customers using a PIN-based debit (ATM) card to initiate transactions (e.g., deposits, withdrawals, account balance inquiries).

Bankcard - A general-purpose credit card, issued by a financial institution under agreement with the bankcard associations (Visa and MasterCard), which customers can use to purchase goods and services and to obtain cash against a line of credit established by the bankcard issuer.

Bilateral Key Security - A multi-level data encryption system, based on the exchange of Bilateral Keys, allowing users of SWIFT to create, send, and receive SWIFT messages. Bilateral Keys are unique authenticator keys possessed by only the two parties (either the provider or recipient of a message) involved and provide confirmation in both directions of the legitimacy of a message sent via SWIFT.

Check - A written order from one party (payer) to another (payee) requiring the payer's financial institution to pay a specified sum on demand to the payee or to a third party specified by the payee

Clearance - The process of transmitting, reconciling, and in some cases, confirming payment orders or financial instrument transfer instructions prior to settlement.

Consumer - Usually refers to an individual engaged in non-commercial transactions.

Correspondent Bank - An institution, acting on behalf of other institutions, that can settle the checks they collect for other institutions (respondents) by using accounts on their books or by sending a wire funds transfers. Generally, a provider of banking and payment services to other financial institutions.

Credit Card - A card indicating the holder has been granted a line of credit. It enables the holder to make purchases or withdraw cash up to a prearranged ceiling. The credit granted can be settled in full by the end of a specified period or can be settled in part, with the balance taken as extended credit. Interest is charged based on the terms of the credit card agreement and the holder is sometimes charged an annual fee.

Currency Balance - As at the time calculated, the current amount (positive or negative) of a particular eligible currency included in an account, as indicated on the books and records of CLS Bank. A currency balance is not a separate account.

Daylight overdraft - A daylight overdraft occurs at any point in the business day when the balance in an institution's account becomes negative. Daylight overdrafts can occur in accounts at Federal Reserve Banks as well as at private financial institutions. Daylight credit can also arise in the form of net debit positions of participants in private payment systems. A daylight overdraft occurs at a Federal Reserve Bank when there are insufficient funds in an institution's Federal Reserve Bank account to cover outgoing funds transfers or incoming book-entry securities transfers. An overdraft can also be the result of other payment activity processed by the Federal Reserve Bank, such as check or automated clearinghouse transactions.

Depository - An institution that holds funds or marketable securities for safekeeping. Depositories may be privately or publicly operated and allow securities transfers through book-entry and offer funds accounts permitting funds transfers as a means of payment.

Electronic Funds Transfer Act (EFTA) - The Electronic Funds Transfer Act and Regulation E are designed to ensure adequate disclosure of basic terms, costs, and rights relating to electronic fund transfer (EFT) services provided to consumers. Institutions offering EFT services must disclose to consumers certain information, including: initial and updated EFT terms, transaction information, periodic statements of activity, the consumer's potential liability for unauthorized transfers, and error resolution rights and procedures. EFT services include automated teller machines, telephone bill payment, point-of-sale transfers in retail stores, fund transfers initiated through the Internet, and pre-authorized transfers to or from a consumer's account.

Encryption - A data security technique used to protect information from unauthorized inspection or alteration. Information is encoded so that data appears as a meaningless string of letters and symbols during delivery or transmission. Upon receipt, the information is decoded using an encryption key.

Extensible Markup Language (XML) - XML (Extensible Markup Language) is a "metalanguage", a language for describing other languages – which lets you design your own customized markup languages for different types of documents. It is designed to improve the functionality of the Web by providing more flexible and adaptable information identification.

Federal Reserve Banks - The Federal Reserve Banks provide a variety of financial services including retail and wholesale payments. The Federal Reserve Bank operates a nationwide system for clearing and settling checks drawn on depository institutions located in all regions of the United States.

Fedwire Funds Service - The Federal Reserve Banks' high-speed electronic funds transfer system. As a real-time gross settlement system, the Fedwire® Funds Service processes and settles individual payments between participants immediately in central bank money. Once processed, these payments are final.

Fedwire Securities Service - The Federal Reserve Banks' high-speed electronic payments system for maintaining securities accounts and for effecting securities transfers. The Fedwire® Securities Service provides a real-time, delivery-versus-payment (DVP), gross settlement system that allows for the immediate, simultaneous transfer of securities against payment. Once processed, securities transfers are final.

FIN (Financial Application) - The SWIFT application within which all SWIFT user-to-user messages are input and output.

Finality - Irrevocable and unconditional transfer of payment during settlement.

Haircut - With respect of an eligible currency, the percentage increase of a negative currency balance or reduction of a positive currency balance and is based on (a) the volatility of the historic foreign exchange movements in the applicable eligible currency determined by CLS Bank and (b) an add-on component.

Instruction - Means (i) any instruction submitted by a Member through the submission process directing CLS Bank to settle certain payment entitlements and obligations arising pursuant to an FX transaction eligible for settlement in CLS Bank and (ii) any instructions resulting from the split of Settlement Eligible Instructions.

Internet - The global system of interconnected computer networks that use the Internet protocol suite (TCP/IP) to link billions of devices worldwide.

Large value funds transfer system - A wholesale payment system used primarily by financial institutions in which large values of funds are transferred between parties. Fedwire® and CHIPS are the two large-value transfer systems in the United States.

Long position - In respect of a currency balance that is greater than zero, the amount by which such currency balance is greater than zero. A position that appreciates in value if market prices increase. When one buys a currency, their position is long.

Matched instructions - Two Instructions in which the information set forth in a specific CLS Bank Rule is matched in accordance with the parameters and procedures set forth in the CLS Bank Rules.

Matching - With respect to compared and non-compared transactions, the process of comparing the trade or settlement details provided by counterparties to ensure they agree with respect to the terms of the transaction. Also called comparison checking.

National Settlement Service (NSS) - Also referred to as Deferred Net Settlement. The Federal Reserve Banks' multilateral settlement service. NSS is offered to depository institutions that settle for participants in clearinghouses, financial exchanges, and other clearing and settlement groups. Settlement agents acting on behalf of those depository institutions electronically submit settlement files to the Federal Reserve Banks. Files are processed on receipt, and entries are automatically posted to the depository institutions' Reserve Bank accounts. Entries are final when posted.

Net debit cap - The maximum dollar amount of uncollateralized daylight overdrafts that an institution is authorized to incur in its Federal Reserve account. The net debit cap is generally equal to an institution's capital times the cap multiple for its cap category.

Office of Foreign Asset Control (OFAC) - The Office of Foreign Assets Control, United States Department of the Treasury, administers and enforces economic sanctions programs primarily against countries and groups of individuals such as terrorists and narcotics traffickers. The sanctions can be either comprehensive or selective, using the blocking of assets and trade restrictions to accomplish foreign policy and national security goals.

Open market operations - The buying and selling of government securities in the open market in order to expand or contract the amount of money in the banking system.

<u>Originating depository financial institution (ODFI)</u> - A participating financial institution that originates entries at the request of and by agreement with its originators in accordance with the provisions of the NACHA rules.

<u>Originator</u> - A person that has authorized an ODFI to transmit a credit or debit entry to the deposit account of a receiver at an RDFI.

<u>Payment</u> - A transfer of value.

<u>Payment system</u> - The mechanism, the rules, institutions, people, markets, and agreements that make the exchange of payments possible.

<u>Payments System Risk Policy (PSR)</u> - The Federal Reserve's Payments System Risk (PSR) policy addressing the risks that payment systems present to the Federal Reserve Banks, the banking system, and to other sectors of the economy.

<u>Real time gross settlement (RTGS) System</u> - A type of payments system operating in real time rather than batch processing mode. It provides immediate finality of transactions. Gross settlement refers to the settlement of each transfer individually rather than netting. FedwireÒ is an example of a real time gross settlement system.

<u>Receiver</u> - An individual, corporation, or other entity that has authorized a company or an originator to initiate a credit or debit entry to a transaction account belonging to the receiver held at its RDFI.

<u>Receiving depository financial institution (RDFI)</u> - Any financial institution qualified to receive debits or credits through its ACH operator in accordance with the ACH rules.

<u>Regulation E</u> - A regulation (12 CFR 205) promulgated by the Board of Governors of the Federal Reserve System to ensure consumers a minimum level of protection in disputes arising from electronic fund transfers.

<u>Reserve account</u> - A non-interest-earning balance account institutions maintain with the Federal Reserve Bank or with a correspondent bank to satisfy the Federal Reserve's reserve requirements. Reserve account balances play a central role in the exchange of funds between depository institutions.

<u>Retail payments</u> - Payments, typically small, made in the goods and services market.

<u>Security procedure agreement</u> - An agreement between a financial institution and a Federal Reserve Bank whereby the financial institution agrees to certain security procedures if it uses an encrypted communications line with access controls for the transmission or receipt of a payment order to or from a Federal Reserve Bank.

<u>Settlement</u> - The final step in the transfer of ownership involving the physical exchange of securities or payment. In a banking transaction, settlement is the process of recording the debit and credit positions of the parties involved in a transfer of funds. In a financial instrument transaction, settlement includes both the transfer of securities by the seller and the payment by the buyer. Settlements can be "gross" or "net." Gross settlement means each transaction is settled individually. Net settlement means parties exchanging payments will offset mutual obligations to deliver identical items (e.g., dollars or EUROS), at a specified time, after which only one net amount of each item is exchanged.

<u>Settlement eligible instructions</u> - See "Matched Instructions".

Short position - In respect of a currency balance that is less than zero, the amount by which such currency balance is less than zero. An investment position that benefits from a decline in market price. When one sells a currency their position is short.

Short position limit - In respect of an eligible currency, the maximum short position a Settlement Member may have at any time in that eligible currency and, unless otherwise reduced pursuant to the CLS Bank Rules, shall equal (i) the total amount of all available committed liquidity facilities in such eligible currency (or such lesser amount that CLS Bank may determine from time to time) minus (ii) the amount of the largest available committed liquidity facility among such liquidity facilities (after taking into account any amounts already drawn.

Spot - The most common foreign exchange transaction. Spot or spot date refers to the spot transaction value date that requires settlement within two business days, subject to value date calculation.

Test key - Internal controls used to verify the authenticity of incoming wire requests involve the use of test keys. A test key is a formula used to develop or interpret test codes or test words. Test codes or words consist of a series of numbers signifying different types of information and usually precede the text of the message. As an example, a test code may contain a bank number, the amount of the transaction, and a number indicating the day and week of the month. As an additional precaution, many test codes contain a variable (sequence number) based on the number of messages received.

Appendix C: Laws, Regulations and Guidance

Laws

- 12 USC: 248 (i), (j), and (o): Federal Reserve Act (N/A)
- 12 USC: 342: Federal Reserve Act (N/A)
- 12 USC: 360: Federal Reserve Act (N/A)
- 12 USC: 464: Federal Reserve Act (N/A)
- 12 USC: 4001-4010: Federal Reserve Act (N/A)

Federal Reserve Board

- 12 CFR Part 210, Subpart B: Funds Transfers Through Fedwire (Regulation J) (N/A)

Appendix D: Legal Framework for Interbank Payment Systems

State and federal statutes, regulations, and case law govern the payment systems in the United States. The relevant legal principles depend on the method of payment (paper-based or electronic) and in some cases the status of parties to a payment (consumer, merchant, or financial institution). Several federal laws apply to payment activities, particularly in the consumer sector. At the state level, the Uniform Commercial Code (UCC) establishes a set of model statutes governing certain commercial and financial activities, including some banking and securities market transactions. Articles of the UCC pertinent to payment and settlement activities are the following: Article 3 (negotiable instruments), Article 4 (bank deposits and collections), Article 4A (funds transfers, including wholesale ACH credit transfers) and Article 8 (investment securities). [15]

Every state has incorporated these Articles, sometimes with local variations, into the state laws. In addition, the rules and membership agreements of private clearing and settlement arrangements provide a contractual framework for payment activity within the relevant governing law.

Fedwire Funds Service

The Federal Reserve's Regulation J governs payment transactions using the Fedwire Funds Service and incorporates the requirements of Article 4A of the UCC. The Regulation, in particular subpart B, defines the rights and responsibilities of financial institutions that use Fedwire, as well as the rights and responsibilities of the Federal Reserve Bank. Federal Reserve Bank Operating Circular 6 covers items such as Fedwire operating hours, security, authentication, fees, and certain restrictions. [16] It also contains time schedules, holidays, and guidelines pertaining to the extension of Fedwire hours.

Under subpart B of Regulation J and Operating Circular 6, the Federal Reserve Banks can impose conditions on an institution's use of Fedwire. In particular, the regulation and operating circular require each Fedwire participant to enter into a security procedure agreement with its Federal Reserve Bank and includes institution responsibilities for information security, business continuity, and related administrative information.

Federal Reserve Regulation CC, Availability of Funds and Collection of Checks, also regulates the time within which a depository institution receiving a Fedwire or CHIPS funds transfer on behalf of a customer must make those funds available to its customer.

National Settlement Service (NSS)

Federal Reserve Bank Operating Circular 12 establishes the terms and conditions under which participants using NSS submit settlement files to the Federal Reserve Banks. The provision of intraday settlement finality is similar to the Fedwire Funds Service and enables the Federal Reserve Banks to manage and limit settlement risk by incorporating risk controls on extensions of daylight credit.

Messaging Systems

Payment messaging systems allow financial institutions to initiate payment orders, providing that the institution is authorized to act on behalf of its customers. It is important for financial institutions to establish the authenticity and time of receipt of payment order messages. UCC4A establishes responsibility for the execution of a payment order and requires financial institutions to agree to the terms and conditions established by the responsible messaging system with respect to security procedures.

CHIPS

Funds transfers made through CHIPS are subject to CHIPS rules and procedures. The CHIPS rules stipulate that the laws of the state of New York, which include Article 4A of the UCC, also apply to CHIPS transactions. CHIP Co. is not responsible for losses resulting from system errors. Each participant agrees to indemnify and to hold harmless CHIPS from such losses. Any participant losses are settled by a loss sharing agreement, and if a participant commits a fraud, that participant will bear the loss. CHIP Co. maintains a financial institution bond for possible employee fraud. Losses exceeding CHIPS's bond coverage are shared on a pro rata basis of each participant's average daily CHIPS usage for the 30-day calendar period preceding the notice of loss to the underwriters of the bond.